Study on bovine mastitis in Chittagong

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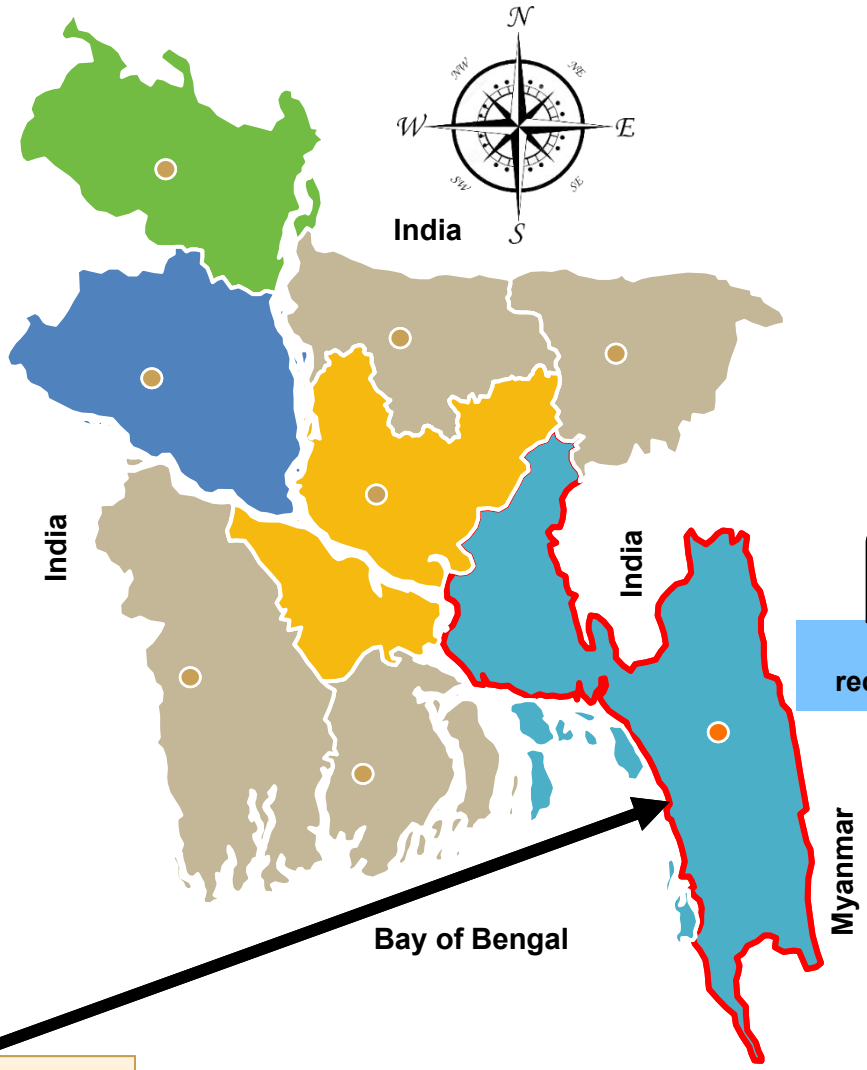
Presentation outlines

Background

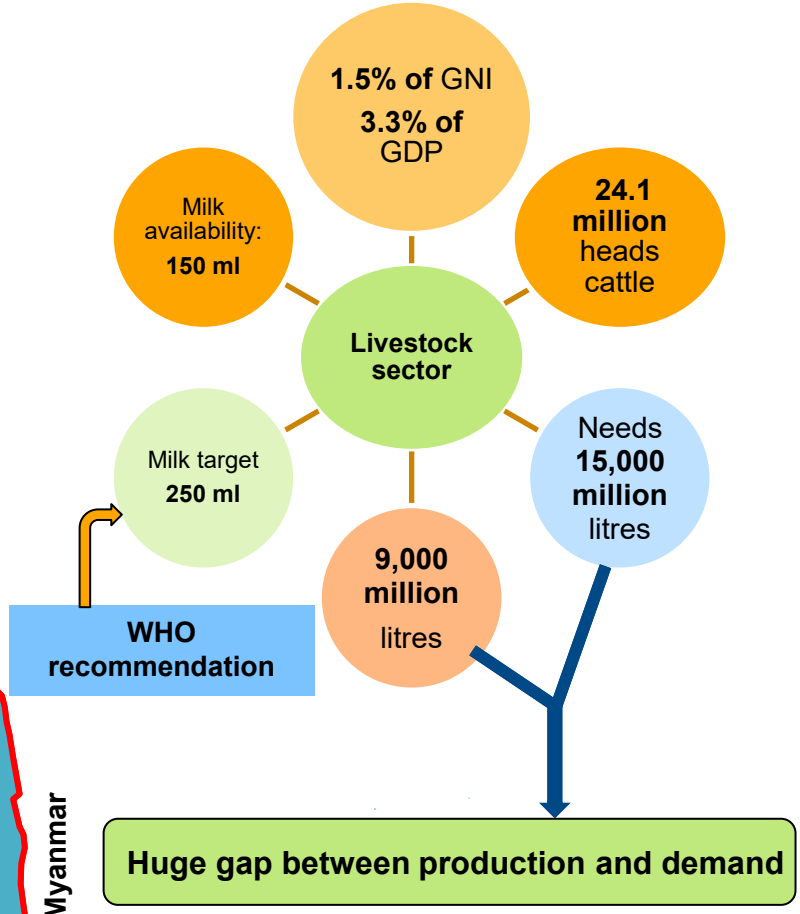
8 divisions

64 districts

Population: 168.0 million
Total area: 1,47,570 sq. Km



Chittagong division



Background

Current drivers in dairy development

To meet the gap between demand and production

Government Initiative

Recently 500 million USD investment for livestock and dairy development.

Involvement into profession

Educated Youth becoming interested in dairy farming

Dairy farms with better management are increasing

Background

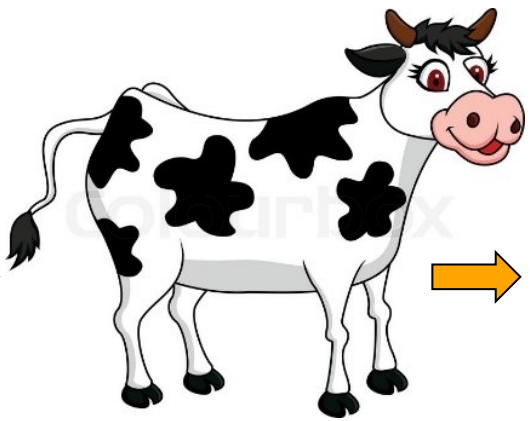
Chittagong is one of the dairy hub in Bangladesh

Inspiration from Elite people

Veterinary care

Govt. support

1,551 dairy farms



2,55,744 heads

Cross breeds

- Local X Holstein Friesian
- Holstein Friesian X Jersey
- Holstein Friesian X Shahiwal

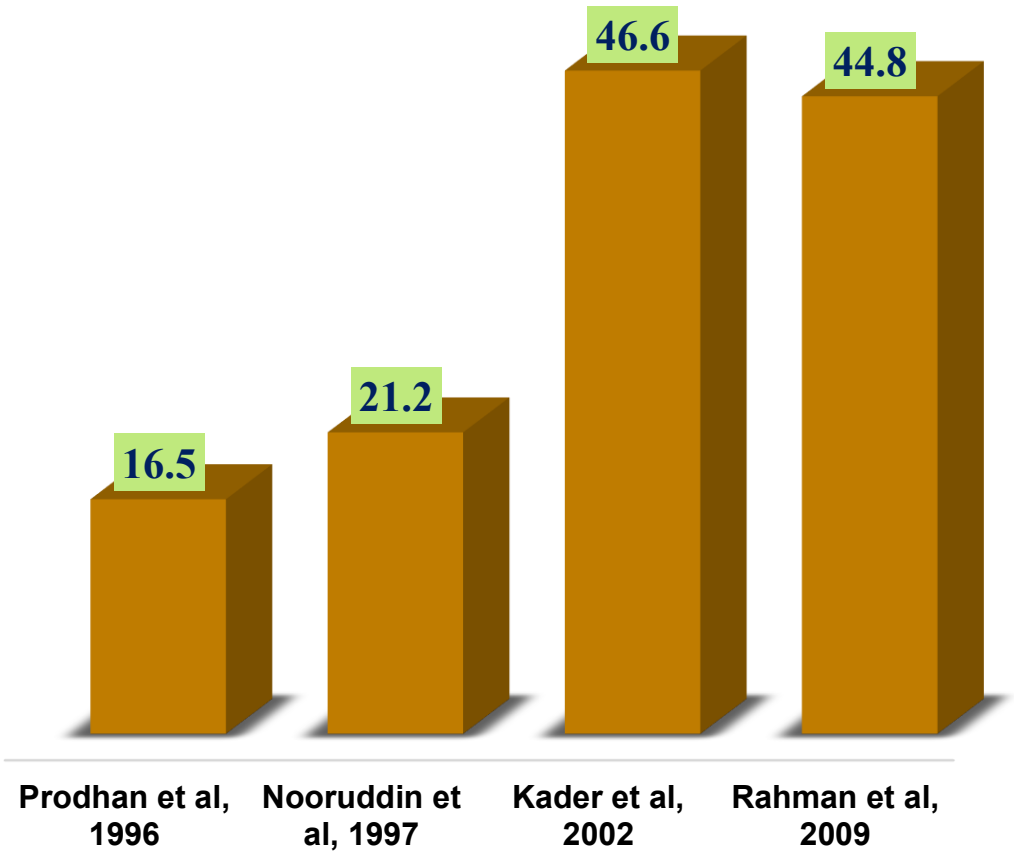
Background

Mastitis is a major challenge for dairying in Bangladesh

Challenges

- Intensive farming system
- Cross bred cows
- Fluctuation in temp. and humidity
- Feed scarcity

Mastitis picture



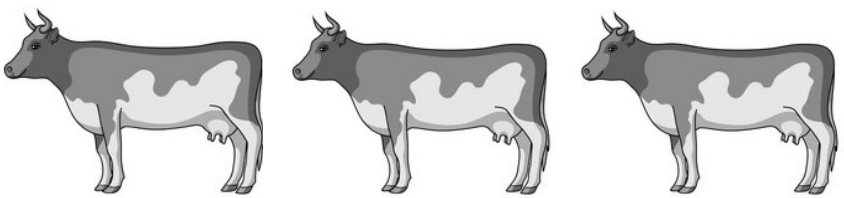
Prevalence of mastitis reports in Bangladesh: Sharma *et al.*, 2012

Objectives

1

Clinical mastitis incidence rate and grading

2

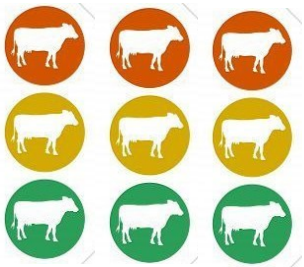


CM associated risk factors

3

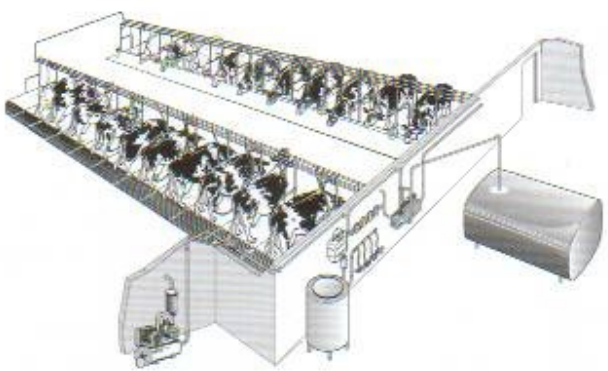
Involved causal agents

Methodology

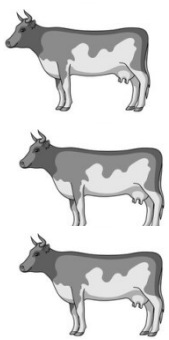


- **Lactating cows**
- Selected farms in **Chittagong** Metropolitoon and surrounding areas

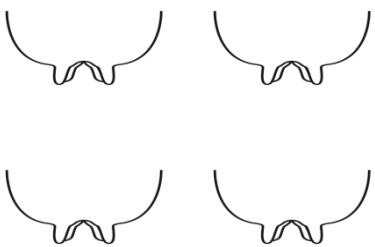
Source population



Farm



Cow

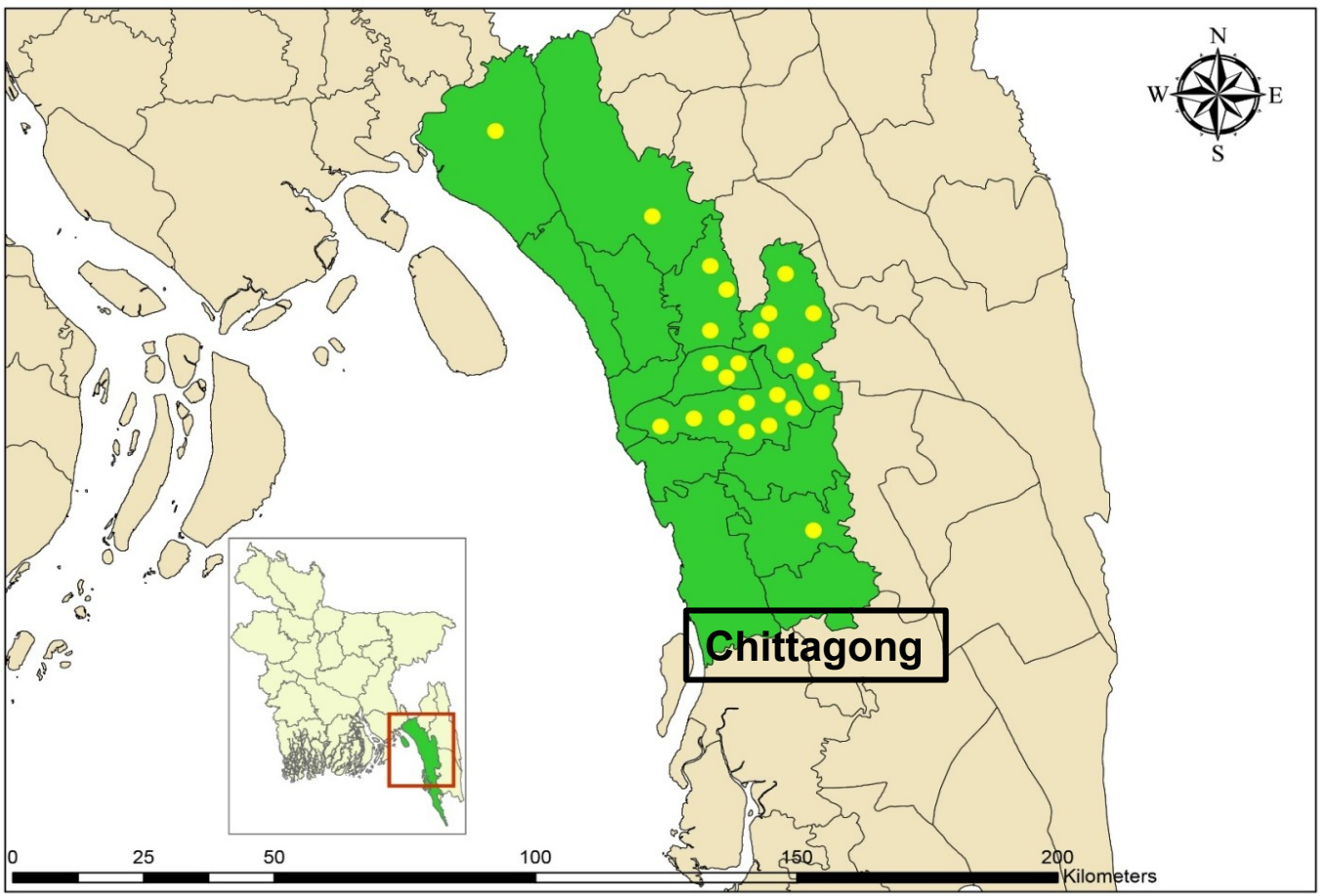


Quarter

Epi unit

Methodology

Location of selected 24 dairy farms under the study



Study duration

6 Months

Methodology

Data Collection

Baseline data

Farm Data
- Population size
- Pregnant/Non-pregnant animals
- Milking animals

Animal Data
- Age, Parity, BCS
- Current Milk Yield
- Lactation Stage
- Days after Parturition

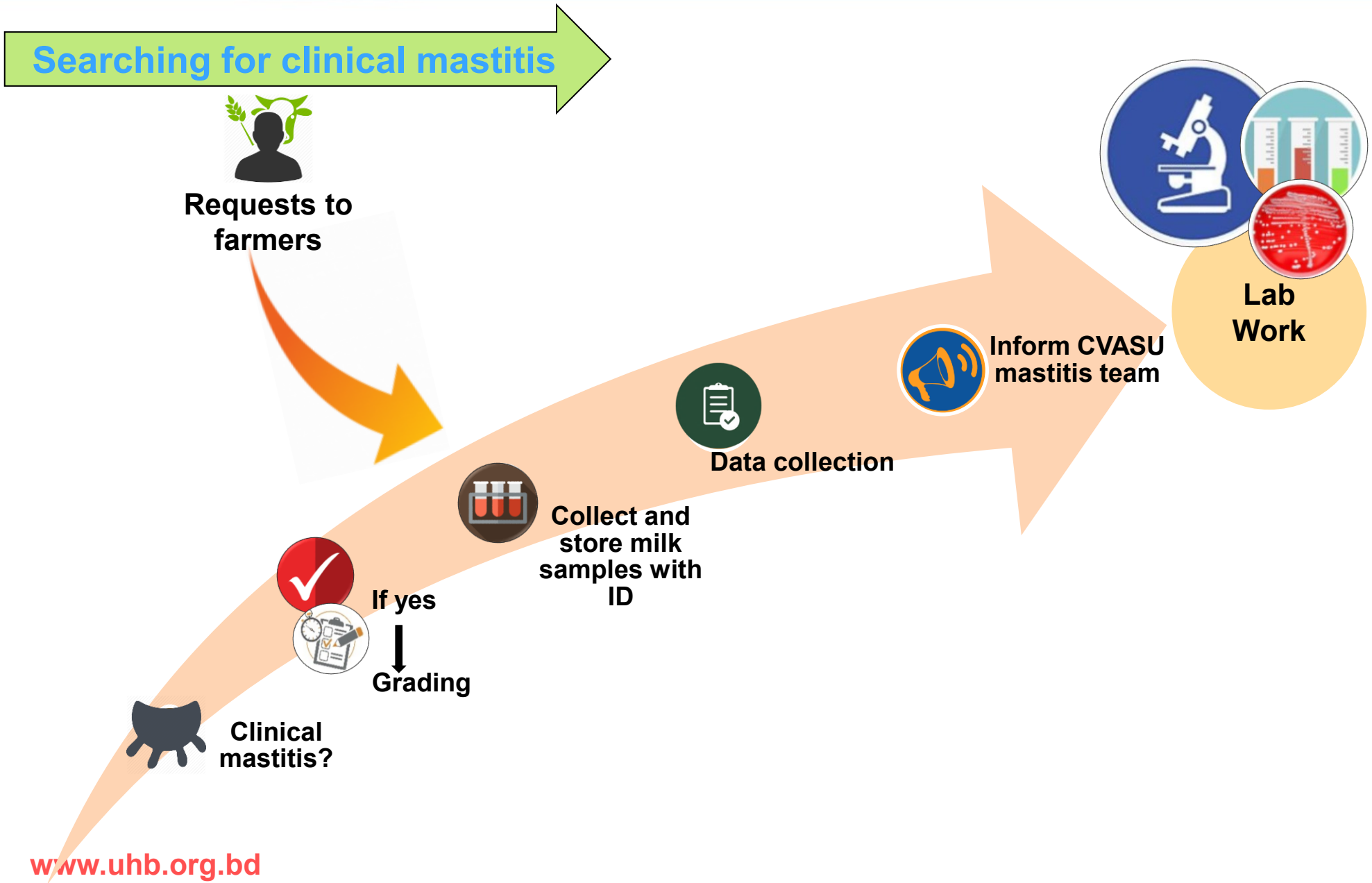
Dynamic data

- Pregnant heifer turned into lactating
- Culled lactating cows
- Newly introduced cows
- Dried cows
- Managemental changes
- Temperature and humidity

1st	2nd	3rd	4th	5th
June	July	August	September	October

Data were collected at one month interval

Methodology



Methodology

Mastitis Grading Record Form for Dairy Farmers

General Information

Date of recording: _____/_____/_____
Cow identification number:
Pregnant 1=Yes; 2=No
Suckled by calf 1=Yes; 2=No
Was the cow treated with antibiotics before the last dry period?
1=Yes; 2=No
If yes, how dry cow therapy is applied?

Observable characteristics of clinical mastitis

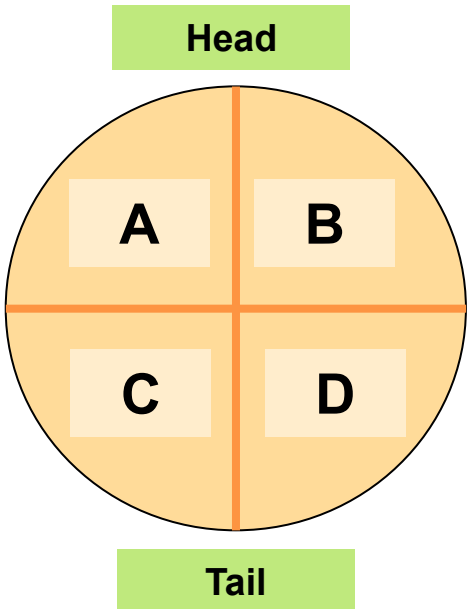
No of quarters affected: _____

Mild (Grade 1): Abnormal milk

Moderate (Grade 2): Abnormal milk accompanied by swelling or redness of mammary gland:

Severe (Grade 3): Exhibited systemic signs of illness such as depression, anorexia, dehydration, or fever:

If any other udder/teat lesions, what are those?



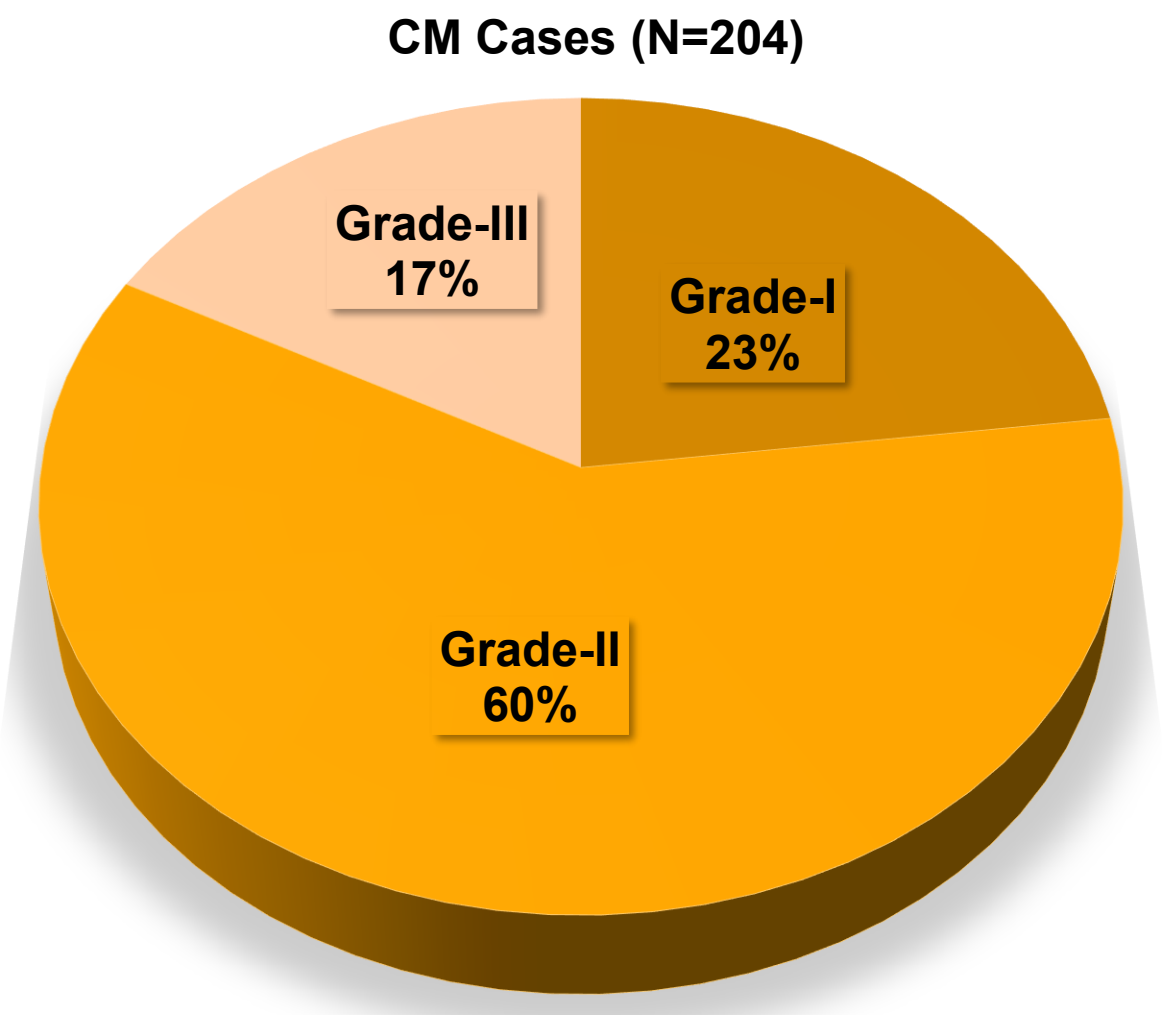
A= Front right
B= Front left
C= Hind right
D= Hind left

Results

1. Incidence rate of Clinical mastitis (CM)
2. CM Factors association
3. Pathogen distribution

Results

Clinical mastitis grading

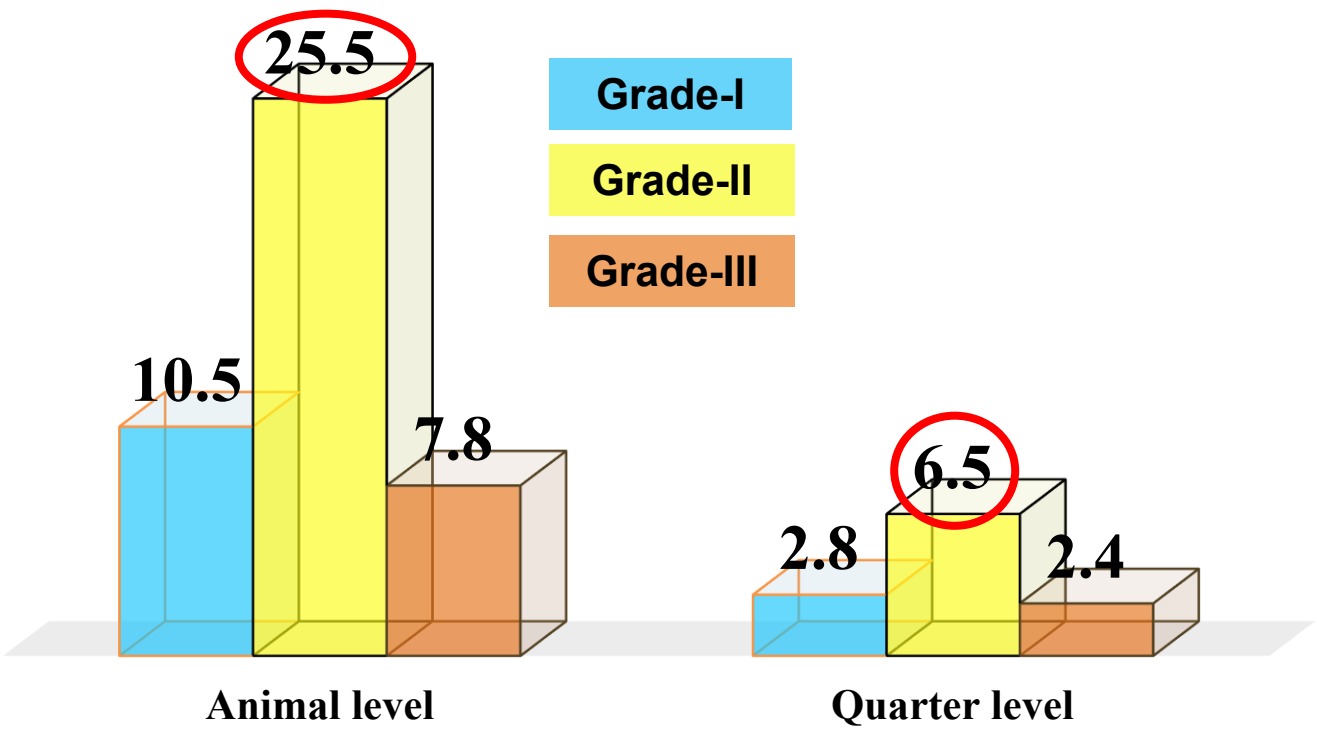


Grade	Number of cases
I	47
II	122
III	35

Results

Clinical mastitis incidence rate

Cow level and quarter level incidence rate at 100 unit/year



Overall

43.9 cases//100 cows/year

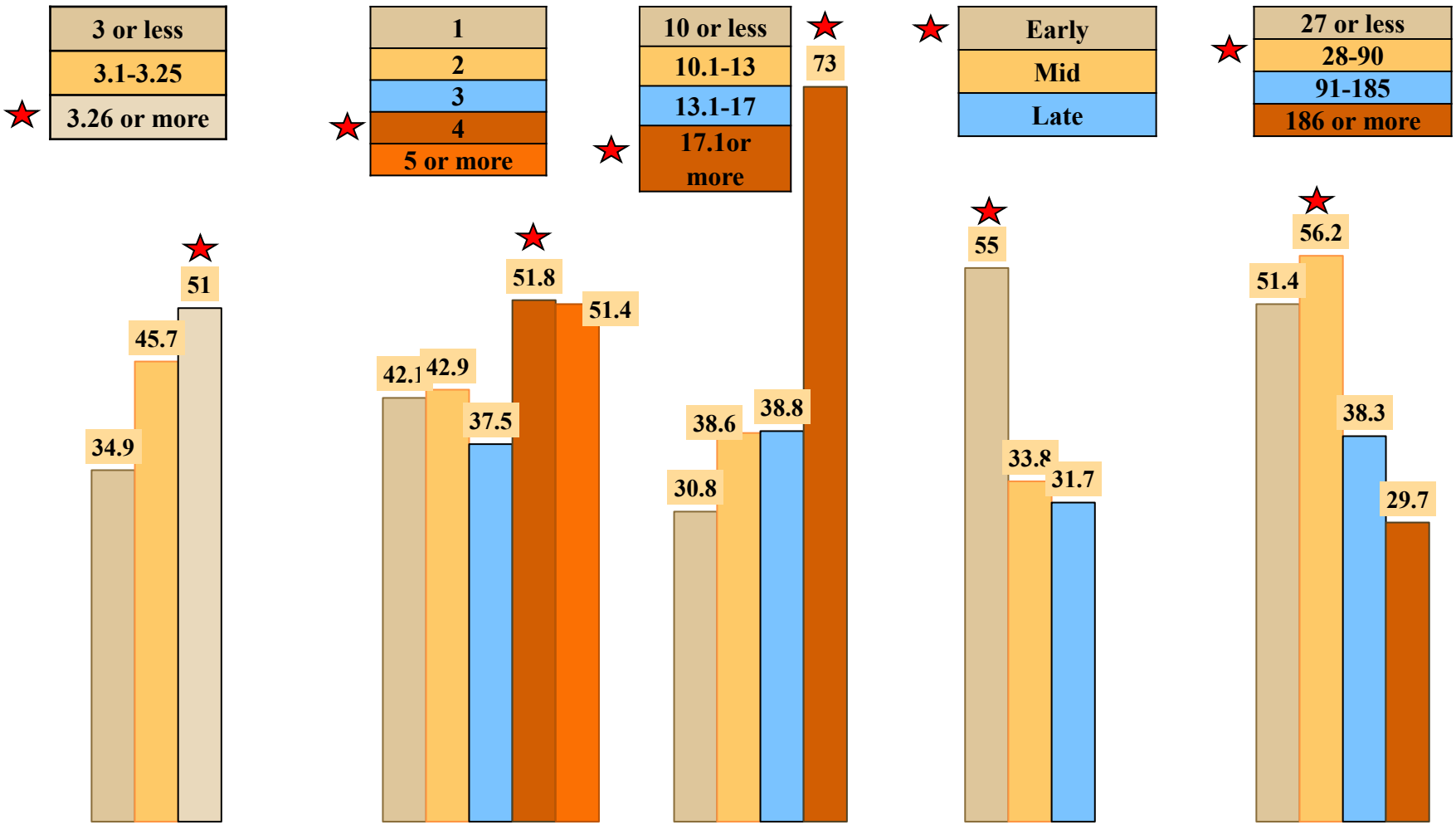
11.6 cases/100 quarters/year

Overall

Results

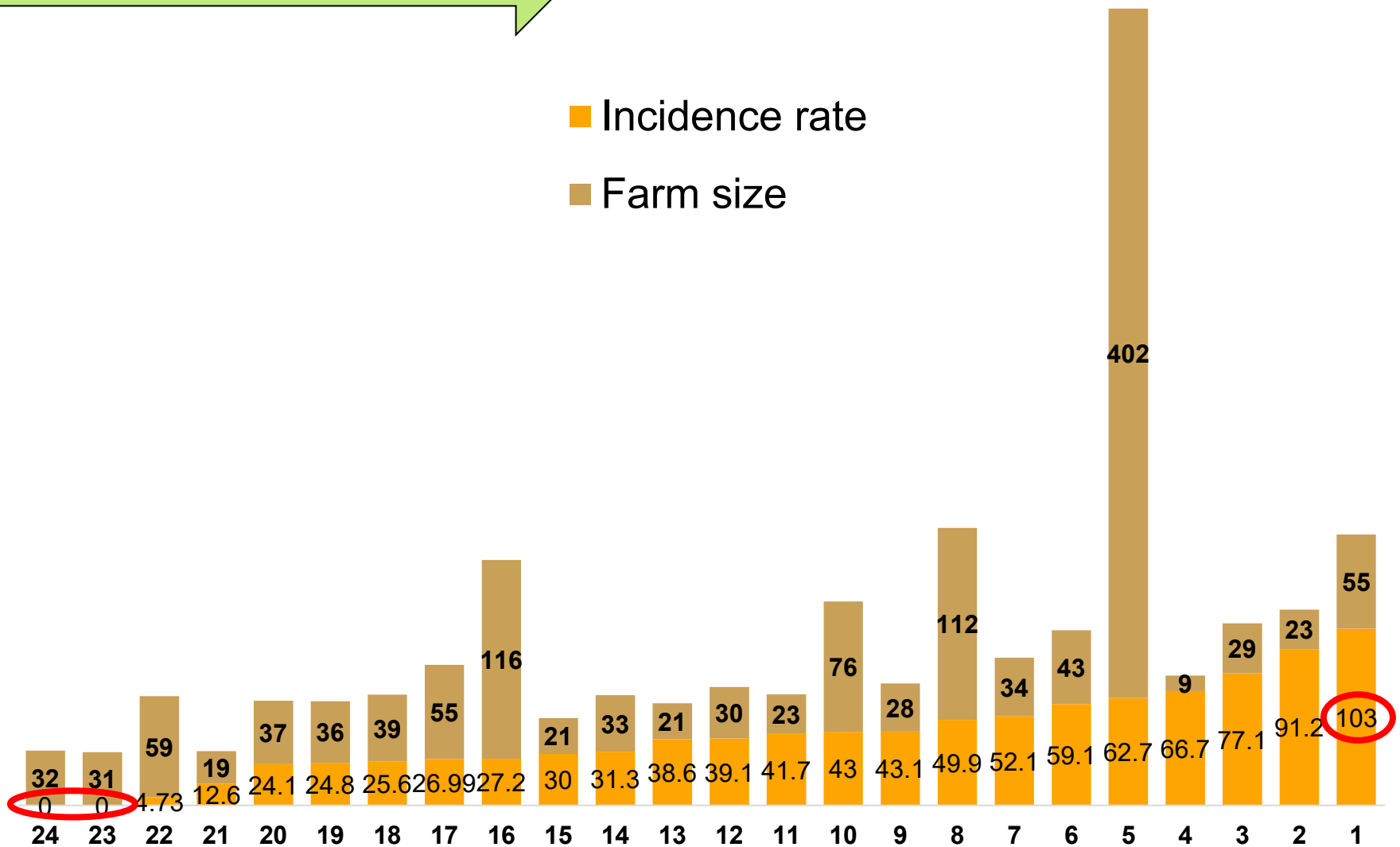
CM incidence rate by factors

Cow level incidence rate at 100 cows/year



Results

Within farm level incidence rate



Results

Animal level risk factors

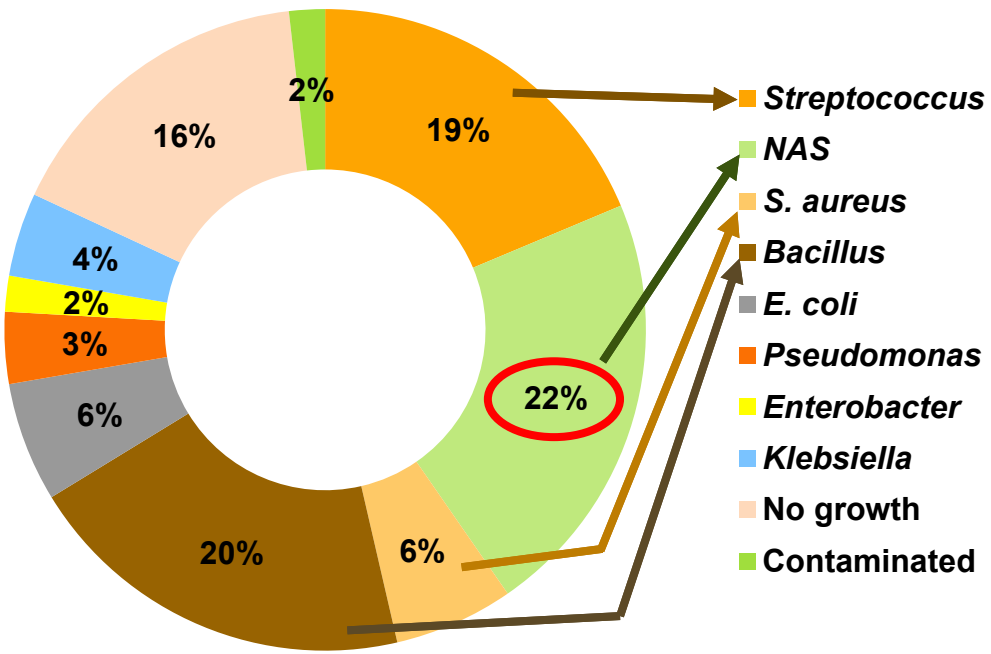
Multivariable logistic regression model using farm as a random effect (N=196 cases of Clinical mastitis)

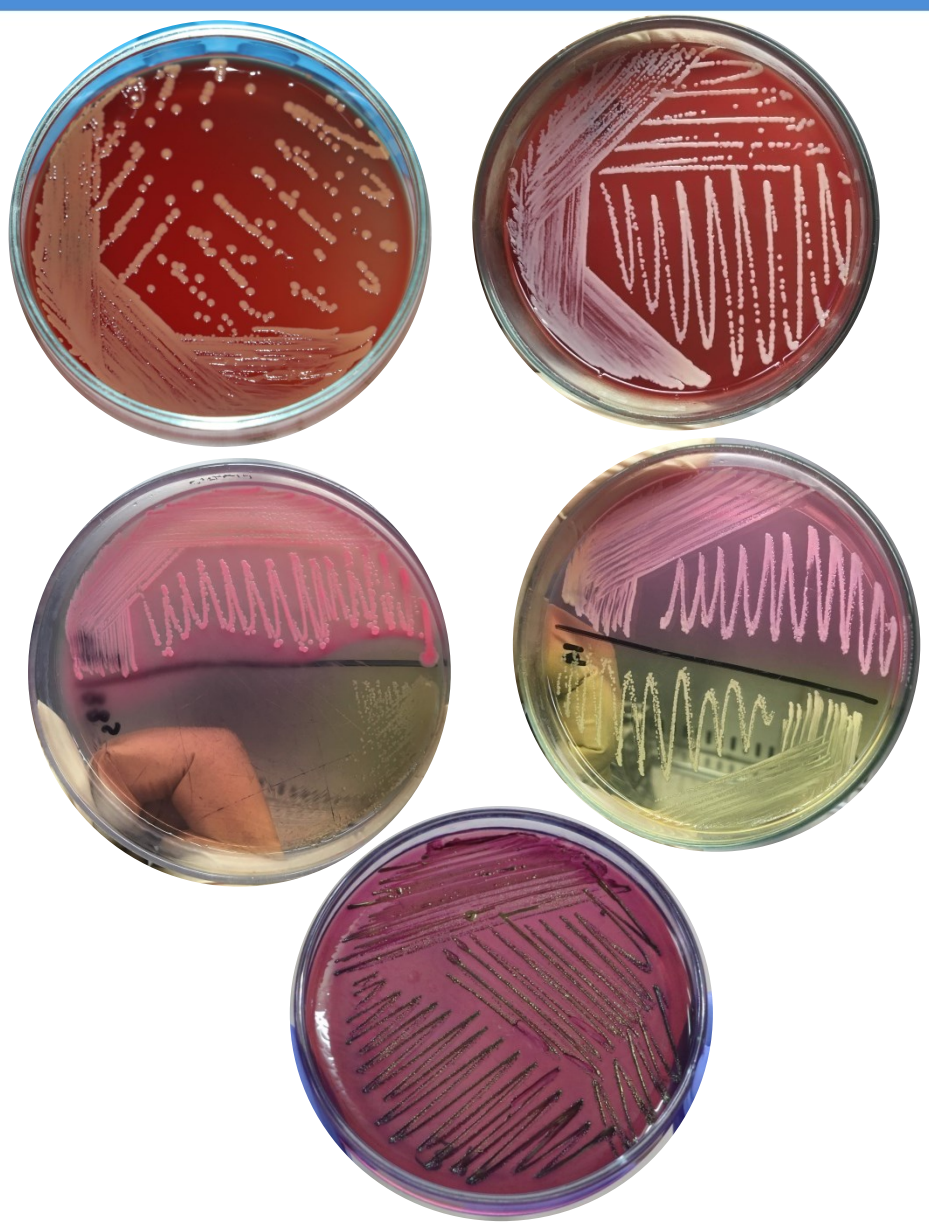
8 animals were excluded from model as they became case at first day of entry into the study

Variable	Categories	N=1383	Odd Ratio	P-value	95% CI
BCS	3 or less	427	Ref		
	3.1-3.25	488	1.2	0.4	0.8-1.8
	3.26 or more	468	1.4	0.1	0.9-2.1
Lactation Stage	Late	352	Ref		
	Mid	283	1.1	0.7	0.7-1.8
	Early	748	1.3	0.2	0.9-2.0
Milk Yield (Liters)	10 or less	462	Ref		
	10.1-13	266	1.3	0.2	0.8-2.1
	13.1-17	328	1.4	0.1	0.9-2.24
	17.1or more	327	2.6	<0.001	1.6-4.0

Results

Frequency distribution (%) of isolated bacteria (n=166)





Conclusions

- High Incidence rate of CM
- Large variation between farms
- Milk yield responsive to CM

Acknowledgements

Thank you for your attention!!

UHB
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Udder Health Bangladesh

“ Healthy Udder
Safe Milk”